Logic Devices

Computer Organization

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logic device

a reusable block, built from gates,

that performs one common task

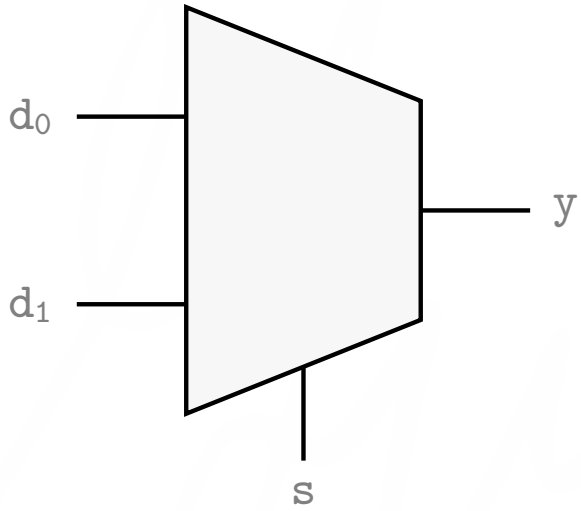
We treat each device as a single symbol and reason about what it does, not how its gates are wired.

- **multiplexer**
  selects one input to pass through
- **demultiplexer**
  selects one output to pass the input through
- **decoder**
  turns a number into a one-hot signal
- **encoder**
  turns a one-hot signal back into a number

These devices give us the vocabulary to describe larger circuits without drawing every gate.

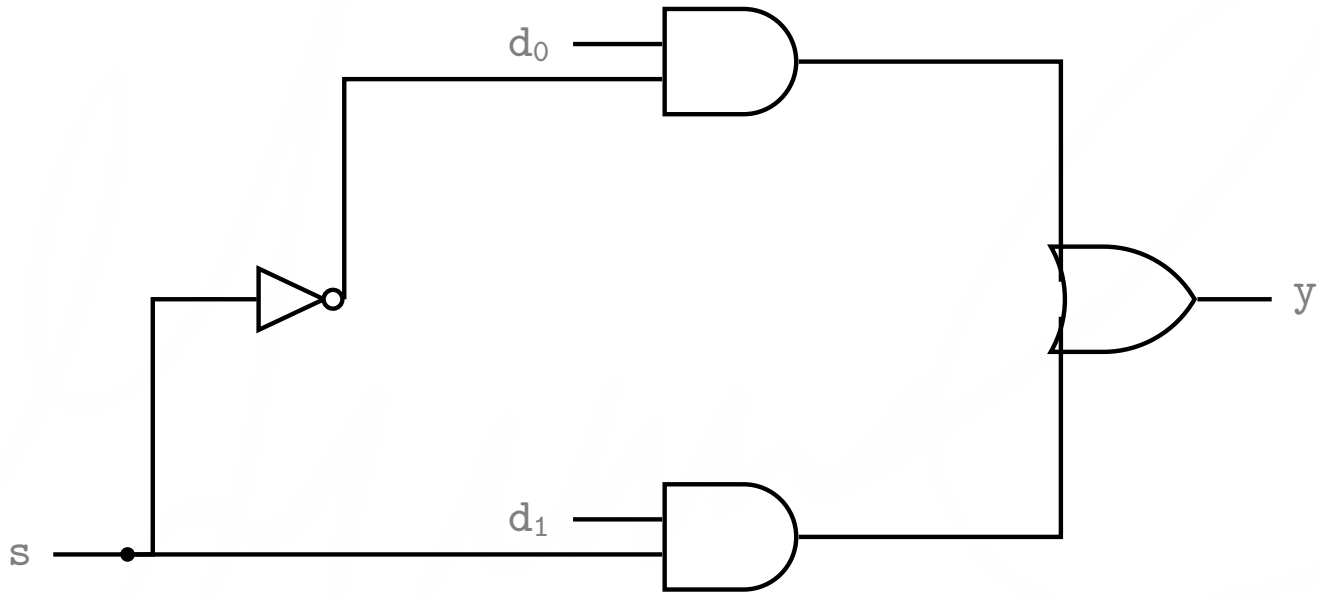
multiplexers

A **multiplexer** passes one of its data inputs to the output. A select input chooses which one.



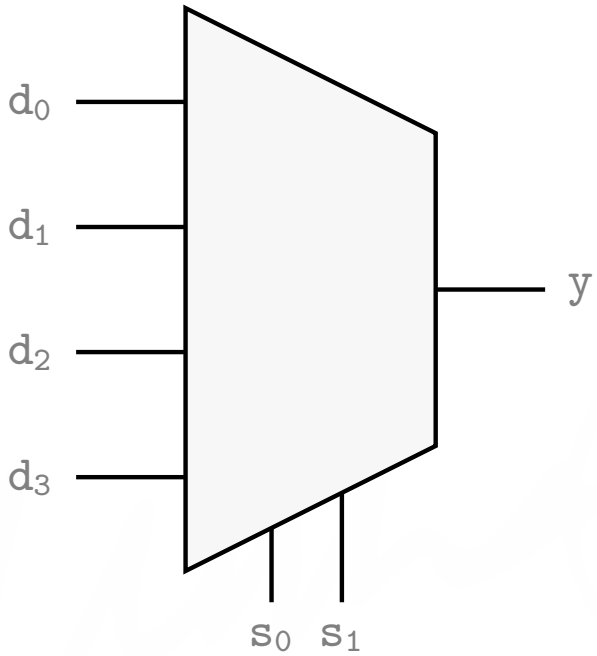
s	y
0	d_0
1	d_1

Inside, the select line either enables one AND gate or the other. Their outputs are combined by an OR.



$$y = \overline{s} \cdot d_0 + s \cdot d_1$$

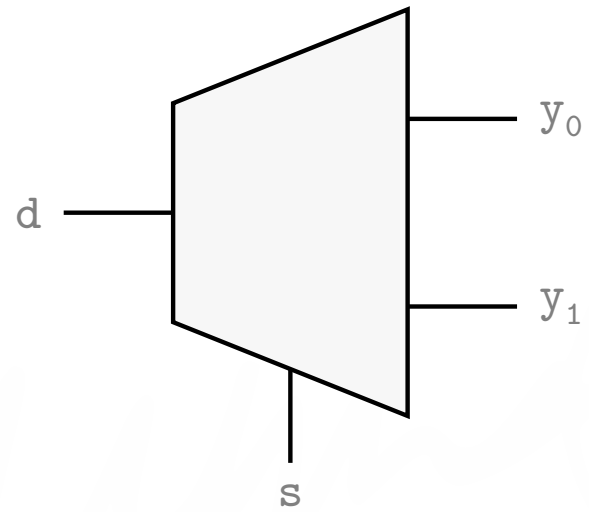
With more select inputs, a multiplexer chooses among more data inputs.



the select inputs read as a number that picks the data input

de-multiplexers

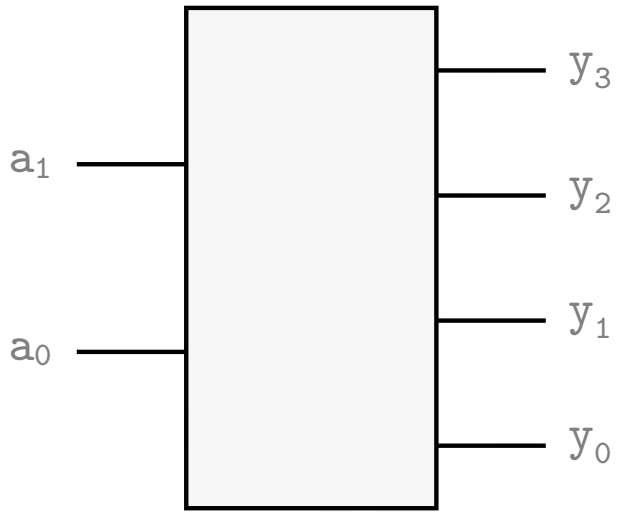
A **de-multiplexer** does the reverse. It sends one data input to the one output chosen by the select line.



a multiplexer merges many lines into one, a demultiplexer splits one into many

decoders

A **decoder** reads its inputs as a number and sets exactly one output to 1.

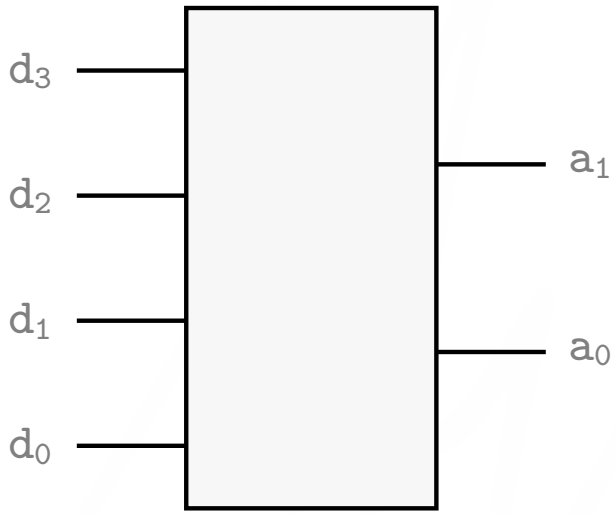


a_1	a_0	y_3	y_2	y_1	y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

This is called a one-hot output.

encoders

An **encoder** does the reverse of a decoder. Given a one-hot input, it outputs the number of the input that is 1.



d_3	d_2	d_1	d_0	a_1	a_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

don't-care values

Real devices, like a decoder for a digit display, have inputs that never occur. We mark their output as a **don't-care**, written `X`.

a_1	a_0	y
0	0	1
0	1	0
1	0	X
1	1	X

When we simplify, we may read each `X` as whichever value makes the groups larger. A don't-care is a free choice, used only when it helps.